

The Effects of Poverty, Ethnic Heterogeneity, and Residential Instability on Property Crime

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Abstract

This study tests whether poverty rate, ethnic heterogeneity, and residential instability statistically significantly affect the property crime rate in U.S. cities of 50,000 or more residents, and whether the resulting risk model can guide security-budget allocation for owners of plant, property, and equipment. A common but unreliable shortcut to judging property-loss risk treats foreign-born population share as a risk signal, a folk assumption contradicted by a century of evidence that immigrants commit crime at lower rates than the native-born; this analysis instead tests three community conditions drawn from social disorganization theory. The analysis joins Federal Bureau of Investigation Table 8 city-level property crime counts for 2022 through 2024 with American Community Survey 5-year estimates to produce 2,182 place-years across 771 cities. Ordinary least squares multiple regression, validated with a variance inflation factor check and stabilized with ridge regression, estimated the association between the three community conditions and property crime before a linear program translated predicted risk into a security-spending mix. All three predictors were statistically significant at $\alpha = 0.05$. Each additional percentage point of poverty was associated with roughly 60 more property crimes per 100,000 residents; each 0.1 increase in the Blau heterogeneity index with roughly 114 more; and each additional percentage point of residential mobility with roughly 28 more. The model explained 17% of the variation, an 8× improvement over a foreign-born-share baseline. The results support a defensive allocation of 50% fortification, 35% surveillance, and 15% alarm, and provide a data-driven foundation for property-risk assessment, with application to three growing Northern Utah municipalities.

Keywords: property crime, social disorganization theory, ethnic heterogeneity, residential instability, security resource allocation, linear programming

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Introduction

To what extent do poverty rate, ethnic heterogeneity, and residential instability affect the property crime rate in U.S. cities with a population of 50,000 or more?

Owners of plant, property, and equipment (PPE) in large U.S. cities must decide where property-loss risk is higher so they can direct security spending efficiently. A common but unreliable shortcut is to treat foreign-born population share as a risk signal, a folk assumption found to be contradicted by a century of evidence showing that immigrants commit crime at lower rates than the native-born.¹

Prior research demonstrated that the association between immigration and crime² is better explained by three community-level mediators rooted in social disorganization theory: poverty, ethnic heterogeneity, and residential instability.³ Testing whether these three conditions statistically significantly affect the FBI property crime rate provides a defensible, data-driven foundation for budget allocation, replacing guesswork with measured community characteristics. For the purpose of this study, we will further examine three growing municipalities in Northern Utah: Layton, Ogden, and Logan.

Property crime, which includes burglary, larceny-theft, and motor vehicle theft, directly damages fixed assets. To examine recent trends of property crime we turn to the FBI Crime Data Explorer, which reports these offenses per 100,000 residents for cities whose agencies submit 12 months of data.⁴ The American Community Survey (ACS) 5-year estimates provide the corresponding community conditions at the place level.

This study joins the latest FBI Table 8 city-level property crime counts between 2022–2024 with ACS Data Profiles (DP02, DP03, DP05) for all incorporated U.S. places with population of 50,000 or more. The dependent variable is the property crime rate per 100,000 residents. The three predictors are poverty rate (ACS DP03), the Blau index of ethnic heterogeneity computed from DP05 race/ethnicity shares, and residential instability measured as the percentage of residents who moved in the past year (ACS DP02).

Arson is classified separately by the FBI because of incomplete reporting and is excluded from the property crime count. For the purposes of this study, violent crime is out of scope and is also excluded.

The analysis proceeds in three stages:

- an ordinary least squares (OLS) multiple linear regression that tests the hypothesis;
- a variance inflation factor (VIF) check confirming predictor independence; and
- a Ridge regression producing risk scores that feed a linear-program security-budget allocation.

¹ (National Academies of Sciences, Engineering, and Medicine, 2015)

² (Stowell, 2007)

³ (Kubrin et al., 2021)

⁴ (Federal Bureau of Investigation, 2026)

Stage 1 decides the hypothesis, where Stages 2 and 3 validate the regression and translate findings into an actionable recommendation.

Null hypothesis (H_0): poverty rate, ethnic heterogeneity, and residential instability do not statistically significantly affect the property crime rate.

Alternate hypothesis (H_1): poverty rate, ethnic heterogeneity, and residential instability statistically significantly affect the property crime rate.

Significance level: $\alpha = 0.05$.

Decision rule: reject H_0 if the overall F-test is significant at $\alpha = 0.05$ or at least one of the three coefficient p-values is less than 0.05.

Methods

Two public U.S. government statistical products supply every modeled field. FBI Table 8 lists counts for burglary, larceny-theft, motor vehicle theft, and arson for each city whose law-enforcement agency reported 12 months of data.⁵ ACS 5-year Data Profiles (DP02, DP03, DP04, DP05) provide population, poverty rate, race/ethnicity shares, percent moved in the last year.⁶ The Census Gazetteer national place file supplies land area for computing population density.⁷ Table 1 summarizes these sources and their roles.

Table 1. Data sources and their roles in the analysis.

Role	Measure	Source	File / API
Target (DV)	Property crime rate per 100,000	FBI Table 8	Crime in the U.S., 2022–2024
Predictor	Poverty rate (%)	ACS DP03	5-year economic profile
Predictor	Blau heterogeneity index	ACS DP05	5-year demographic profile
Predictor	% moved in last year	ACS DP02	5-year social profile
Filter	Population $\geq 50,000$	ACS DP05	Total population estimate
Control	Land area (sq mi)	Gazetteer	National place file

Using published government statistical products is reproducible and transparent. Any researcher can download the same FBI Table 8 editions and ACS Data Profiles from public portals, apply the same population and 12-month filters, and arrive at the same place-years. The data is collected under standardized federal definitions; the FBI Uniform Crime Reporting program governs crime counts, and the Census Bureau's probability-based survey design governs community characteristics. This standardization supports comparability across cities and years.

⁵ (Federal Bureau of Investigation, 2019)

⁶ (United States Census Bureau, n.d.-a)

⁷ (United States Census Bureau, n.d.-b)

Police-reported property crime substantially understates the true volume of offenses. The National Crime Victimization Survey indicates that only about one-third of property crimes are reported to law enforcement.⁸ Additionally, the 2021 transition from the Summary Reporting System (SRS) to the National Incident-Based Reporting System (NIBRS) caused many agencies to miss a full year of reporting;⁹ although participation recovered to approximately 94% of the U.S. population by 2023, city-level gaps persist. The study mitigates this by restricting to Table 8's 12-month city publication, which already enforces a coverage floor, and by stacking three years (2022–2024) to increase sample size.

The primary challenge involves matching FBI and Census identifiers. Table 8 lists cities by state name and city name (for example, Ogden under Utah) with no Federal Information Processing Standards (FIPS) code, whereas ACS tables use a GEO_ID and a formatted name (e.g., "Ogden city, Utah"). This study builds a standardized match key by converting state names to postal abbreviations and stripping suffixes such as "city," "town," and "village" from both sides. Unmatched Table 8 entries (mostly townships and multi-agency jurisdictions) and population-ratio mismatches (Table 8 population falling outside the 0.50–1.35 ratio window) are logged in an exclusion file rather than silently dropped, preserving an audit trail of excluded records across the three study years.

This study performs all extraction and preparation in Python 3 within Jupyter notebooks, using pandas for tabular manipulation,¹⁰ NumPy for array computation,¹¹ and matplotlib¹² and seaborn¹³ for visualization. The process follows an extract–transform–load (ETL) sequence and an assess–mitigate–re-assess cleaning cycle, documented across six notebooks.

This study reads the FBI Table 8 CSV files for 2022, 2023, and 2024 into pandas, skipping the published title rows, and forward-filled the State column because Table 8 lists each state name once as a header. The analysis retained the City, Population, Property crime (total), Burglary, Larceny-theft, Motor vehicle theft, and Arson columns, added a year column, and stacked the three years into a single table.

The stacked table contained 25,228 rows: 7,876 in 2022, 8,365 in 2023, and 8,987 in 2024.

Next, we load the ACS 5-year Data Profiles (DP02, DP03, DP05) and the B05002 foreign-born table for each year. Each table contains approximately 31,900–32,000 place rows

⁸ (Bureau of Justice Statistics, n.d.)

⁹ (Kemper, 2025)

¹⁰ (Pandas development team, 2024)

¹¹ (NumPy Developers, 2024)

¹² (The Matplotlib development team, 2024)

¹³ (Waskom, 2024)

nationwide. The GEO_ID and NAME columns serve as the join keys. DP05 supplies total population and race/ethnicity estimates; DP03 supplies poverty rate, unemployment, and median household income; DP02 supplies percent moved in the last year and foreign-born share. Of these, 864–876 places per year had a population of 50,000 or greater.

The Census Gazetteer national place file provides GEOID, land area in square miles (ALAND_SQMI), latitude, and longitude. The analysis confirmed one row per GEOID (~32,200 rows per year) and used land area to compute population density (population / ALAND_SQMI).

This study constructs a standardized match key by (a) mapping Table 8 state names to two-letter postal abbreviations, (b) stripping place suffixes ("city," "town," "village," "CDP") and punctuation from both the Table 8 city name and the ACS NAME field, and (c) upper-casing both sides. An alias dictionary handles consolidated governments (e.g., Nashville-Davidson, Louisville/Jefferson, Indianapolis). The ACS tables are left-joined to Table 8 on state abbreviation + cleaned city name + year. The Gazetteer is then left-joined on ACS GEOID.

The join matched 89.8% of place-years overall (91.0% in 2022, 90.1% in 2023, and 88.5% in 2024).

Place-years are retained only if the ACS total population was 50,000 or more and the place was an incorporated city (not a Census-designated place). A population-ratio guard requires Table 8 population / ACS population to fall within [0.50, 1.35], preventing countywide agencies from pairing with a smaller city denominator.

After the population floor and the population-ratio guard, 2,183 place-years remain across 771 unique places: 706 in 2022, 732 in 2023, and 745 in 2024. Eighteen census-designated places survive because their Table 8 name matched directly. One duplicate GEOID-year pair will be resolved during mitigation.

FBI Table 8 is defined as the 12-month city publication; it does not include a months-reported column because inclusion in the table implies full-year reporting. This study drops rows with missing burglary, larceny-theft, or motor vehicle theft, and performs no imputation of the target variable.

No rows are dropped for missing offense counts. All 2,183 place-years carry complete burglary, larceny-theft, and motor vehicle theft values.

This study computes the following variables from the joined table:

- Property crime count = burglary + larceny-theft + motor vehicle theft (arson excluded).
- Property crime rate = count / ACS population × 100,000.
- Poverty rate = ACS DP03 percent below the federal poverty line.
- Blau heterogeneity index = $1 - \sum p^2$, where each p is a group's share of the group-sum denominator, computed from a set of eight mutually exclusive DP05 race/ethnicity groups: Hispanic (any race), non-Hispanic White alone, non-Hispanic Black alone, non-Hispanic American Indian/Alaska Native alone, non-Hispanic Asian alone, non-Hispanic Native Hawaiian/Pacific Islander alone, non-Hispanic Some Other Race alone, and non-Hispanic Two or More Races.¹⁴

¹⁴ (Blau, 1977)

- Residential instability = percent who lived in a different house one year earlier (ACS DP02).
- Controls: unemployment rate, median household income, population, density, renter share, foreign-born share

`place_year_joined.csv` is the covered universe after the official join, the population filter, and the complete-count rule. Duplicate `geoid-year` keys and rows missing a locked predictor have not yet been dropped. Writing the file at this point separates the extract from the cleaning sequence.

This study standardized column names, reviewed summary statistics in small column groups, ran duplicate checks on `geoid + year`, and tabulated missingness by column; only arson had nulls (87), and all four key columns were fully populated. The analysis checked for impossible values, including negative counts or rates, poverty/renter/moved shares outside 0–100, Blau values outside 0–1, and density ≤ 0 , and found none. It also flagged outliers on the three predictors and the rate using the IQR method (property crime rate: 58 flags; poverty: 17; Blau: 17; mobility: 74) but did not remove them, since high-crime cities are part of what the study examines.

The analysis found one exact duplicate key (Union City, NJ, 2023, where two Table 8 agencies matched the same ACS GEOID), dropped the smaller-population agency, and logged it (2,183 to 2,182 rows). It dropped rows with missing target or missing any predictor, with zero additional rows affected, and enforced data types (year as integer, rates as float). The study applies no winsorization or PCA, because capping extreme values would have erased the very high-crime cities under investigation, and collapsing the three theory-named predictors into composite components would have destroyed the one-to-one coefficient reading the hypothesis test requires.

The analysis repeated the assess block on the mitigated table to confirm zero duplicates, zero nulls in the target and predictors, and all values within valid ranges. It saved the final cleaned table (2,182 rows) and the pre-drop joined table for audit.

Python's pandas library provides vectorized operations that process the entire 2,182-row join and cleaning pipeline in seconds, with built-in functions for merging, grouping, null detection, and file output that make each step auditable. NumPy supplied the array arithmetic behind the derived measures, computing the property crime rate, the Blau index, and population density with the same vectorized speed. Matplotlib and seaborn rendered the exploratory figures directly from the cleaned table, so every chart stays tied to the exact data it summarizes rather than a separately maintained copy.

Pandas loads the entire dataset into memory, which could become a constraint if the study were expanded to all 30,000+ Census places or to incident-level NIBRS microdata; for this study's 2,182 rows, memory was not a practical limitation. NumPy operates on raw numeric arrays and offers no built-in handling of labels or missing-value flags, so the cleaning logic still had to live in pandas. Matplotlib and seaborn produce static images that require explicit code for layout and styling, and they offer no interactive drill-down, which a business owner exploring the data would otherwise want.

To test the hypothesis, the study fit ordinary least squares (OLS) multiple linear regression using Python's statsmodels library.¹⁵ The dependent variable was the property crime rate per 100,000, and the three predictors were poverty rate, Blau ethnic heterogeneity, and percent moved in the last year. The analysis fitted OLS on the full cleaned table of 2,182 place-years, and statsmodels returned the coefficients, standard errors, t-statistics, p-values, R^2 , and the overall F-statistic with its p-value in a single summary.

OLS was selected because the hypothesis names three continuous predictors and one continuous outcome and requires a p-value at $\alpha = 0.05$. It is the standard estimator for this design, returning the coefficients, standard errors, p-values, and the overall F-statistic directly, and it is interpretable, because each coefficient quantifies the expected change in the crime rate per unit change in a predictor, holding the others constant.

OLS assumes homoscedastic, independent residuals. As the diagnostics reported in the Results section show, the residuals are heteroscedastic and the Durbin–Watson statistic indicates positive autocorrelation from the panel structure. These violations can make the default standard errors look more precise than they truly are; clustering the standard errors by city would be more conservative. The primary analysis keeps the default OLS standard errors because the technique under evaluation is the plain regression itself.

As a robustness check, re-estimating the model with standard errors clustered by place still leaves all three coefficients significant (poverty $p < 0.001$, Blau $p < 0.001$, mobility $p < 0.001$), so the hypothesis decision does not depend on the clustering choice.

After fitting OLS, the study computed the variance inflation factor for each predictor using the statsmodels library. VIF measures how much the variance of a coefficient is inflated by collinearity with the other predictors, and the pre-declared rule flags any VIF exceeding 5.

The check was needed because poverty, heterogeneity, and residential instability are plausibly correlated in urban settings, so collinearity should be ruled out before interpreting the OLS coefficients as stable, separate effects.

VIF is the standard one-number diagnostic for multicollinearity; it is fast to compute and has a widely accepted threshold, making it straightforward to evaluate.

VIF detects only linear collinearity among the predictors. If two predictors are related in a nonlinear way (e.g., a U-shaped association), VIF could still be low while the coefficients are affected. In this study, the pairwise Pearson correlations among the three predictors are all below 0.37, so nonlinear overlap is unlikely to be severe.

The study fit ridge regression (ℓ_2 -regularized linear regression) to produce stabilized risk scores for the budget allocation. Ridge shrinks coefficients toward zero by adding a penalty proportional to the sum of squared coefficients, preventing overfitting when predictors are correlated.¹⁶ Unlike Lasso (ℓ_1), ridge retains all three predictors, which is appropriate because the hypothesis requires all three to remain in the model. The analysis fitted the model using

¹⁵ (Perktold et al., 2023)

¹⁶ (Hoerl & Kennard, 1970)

scikit-learn's RidgeCV on a geographic training split (seven of nine Census divisions; $n = 684$ places, 1,939 place-years). The holdout contained the remaining two divisions (New England and Middle Atlantic; $n = 87$ places, 243 place-years). The analysis standardized predictors before fitting.

Ridge keeps all three predictors in the model, producing a fully interpretable risk score that includes every predictor from the hypothesis. It shrinks unstable coefficients under collinearity, yielding a lower-variance prediction than OLS for budget allocation purposes.

Ridge introduces a tuning parameter (λ) whose optimal value depends on the cross-validation scheme. With only 87 holdout places in two adjacent Census divisions, the holdout evaluation has limited geographic diversity, and RMSE estimates carry wide uncertainty.

After producing Ridge risk scores for each place-year, a linear program allocated a planning budget of \$10 million across four CPTED-aligned security control families¹⁷ (fortification, surveillance, alarm, and authorization) and three risk tiers (high: top 10%, mid: next 15%, low: remaining 75%) cut from the predicted rate distribution.¹⁸

The \$10 million figure is a round-number planning pot with no tie to any specific business's revenue or asset value; it exists only to give the solver a concrete total against which to express the percentage mix and compare alternatives. Because every constraint and the objective are stated as shares of that total, the resulting percentage mix is scale-invariant, meaning any budget size produces the same split.¹⁹ The study formulated and solved the linear program using the PuLP library.²⁰

The decision variables $x(g,k) \geq 0$ represent the dollars of control k assigned to risk tier g , where $k \in \{\text{fortification, surveillance, alarm, authorization}\}$ and $g \in \{\text{high, mid, low}\}$, for twelve decision variables in total.

Exposure by tier was 1,182,439 for the high tier, 3,367,544 for the mid tier, and 4,675,129 for the low tier, or 9,225,112 in total.

The objective minimized expected residual loss, $\sum [E(g) - \sum \alpha(k) \cdot x(g,k)]$ across all tiers, where $E(g)$ is the exposure (Ridge-predicted rate $\times$ population / 100,000) for tier g and $\alpha(k)$ is the effectiveness per dollar for control k .²¹

The constraints specified: (1) total budget, $\sum x(g,k) \leq \$10,000,000$; (2) residual floor, with residual not falling below 50% of exposure in any tier;²² (3) per-control ceiling, with no family taking more than 50% of budget; (4) per-tier ceiling, with no tier taking more than 50% of

¹⁷ (Cozens et al., 2005)

¹⁸ (Postek et al., 2022)

¹⁹ (Kallrath, 2021)

²⁰ (Mitchell et al., 2009)

²¹ (Williams, 2013)

²² (Lord & Nunes-Vaz, 2013)

budget; (5) high-tier floor of at least 25% of budget; (6) prevention (fortification + surveillance) of at least 60% of budget, with fortification and surveillance each at least 20%; (7) alarm between 15% and 25%; and (8) preventive dollars and the alarm floor placed in the high and mid tiers.²³

The effectiveness coefficients (α_k) were derived from a security-price book compiled from public vendor and CPTED sources, finalized before the solver ran. Because published loss-reduction figures were sparse, all four families used a conservative 0.20 effectiveness fallback.

The linear program provides a globally optimal allocation under the stated constraints, replacing equal-share guesswork with a principled, reproducible recommendation. The formulation is transparent, with every constraint and coefficient stated before the solver runs.

The linear program assumes linear effectiveness, meaning that each additional dollar of a control family reduces the same amount of expected loss, and the $\alpha(k)$ values are derived from limited published literature (all four families defaulted to 0.20 effectiveness). If the true relationship between spending and crime reduction is nonlinear or interactive across families, the optimal mix could differ substantially.

Results

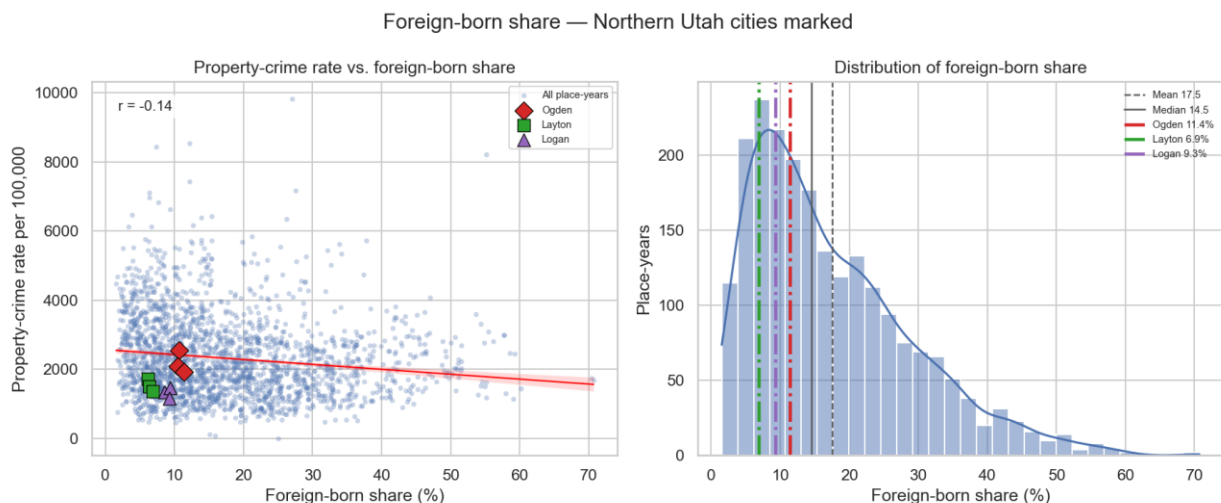


Figure 1. Foreign-born share vs. property crime rate and distribution, with Northern Utah cities marked.

A common but unreliable shortcut treats foreign-born population share as a proxy for property-crime risk. Figure 1 shows a weak negative association ($r \approx -0.14$): cities with a higher foreign-born share tend to have lower property-crime rates, not higher, and all three Northern Utah focus cities sit below the national mean of 17.5%.

A simple linear regression of the property crime rate on foreign-born share confirmed the association, returning a coefficient of -14.21 crimes per 100,000 per percentage point. The

²³ (Bradley et al., 1977)

t-statistic of -6.76 expresses that coefficient in standard-error units, meaning the slope sits more than six standard errors below zero; the corresponding p-value of less than 0.001 reports the probability of observing a slope this far from zero by chance if foreign-born share truly had no effect, a probability so small that the negative relationship is statistically significant.

Although the slope is significant, it points the opposite direction of the folk assumption and explains only 2.1% of the variation ($R^2 = 0.021$), so a signal this weak cannot guide a spending decision; the regression stages that followed tested a theory-grounded alternative.

Before fitting the multiple linear regression, exploratory analysis characterized the distributions and bivariate relationships among the three theory-grounded predictors and the property crime rate in the cleaned table (Figures 2–4).

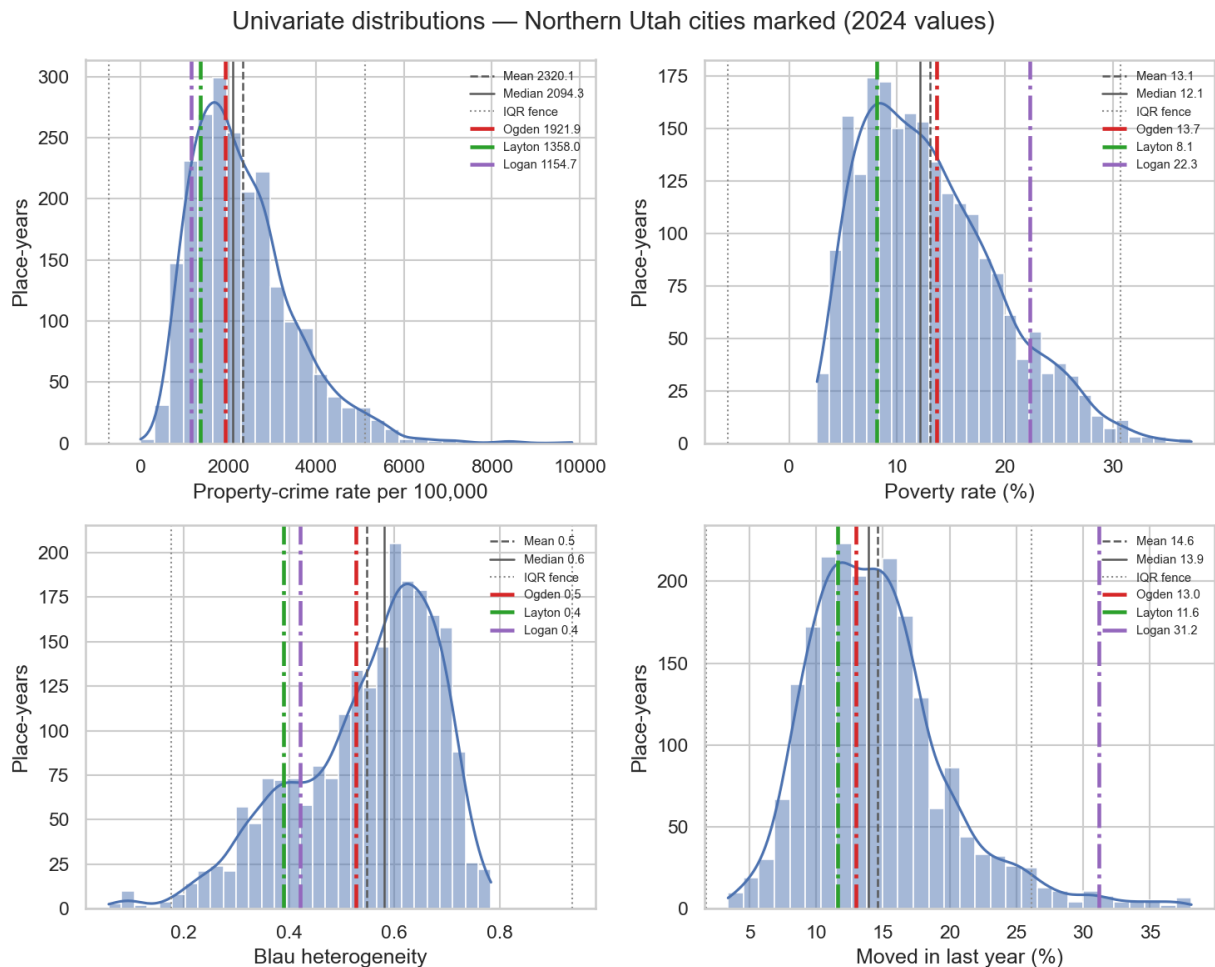


Figure 2. Univariate distributions of the property crime rate and three predictors.

The property crime rate is right-skewed, with a median of approximately 2,094 per 100,000 and a mean of approximately 2,320 per 100,000. Most cities cluster below 3,000, with a long tail extending past 8,000. Poverty rate (mean 13.1%, median 12.2%) and percent moved (mean 14.6%, median 13.9%) are also right-skewed, with their means pulled above their medians by a few cities at the high end.

Blau heterogeneity (mean 0.548, median 0.580) is instead slightly left-skewed, its mean falling below its median because most cities bunch toward the more heterogeneous portion of the 0-to-1 scale.

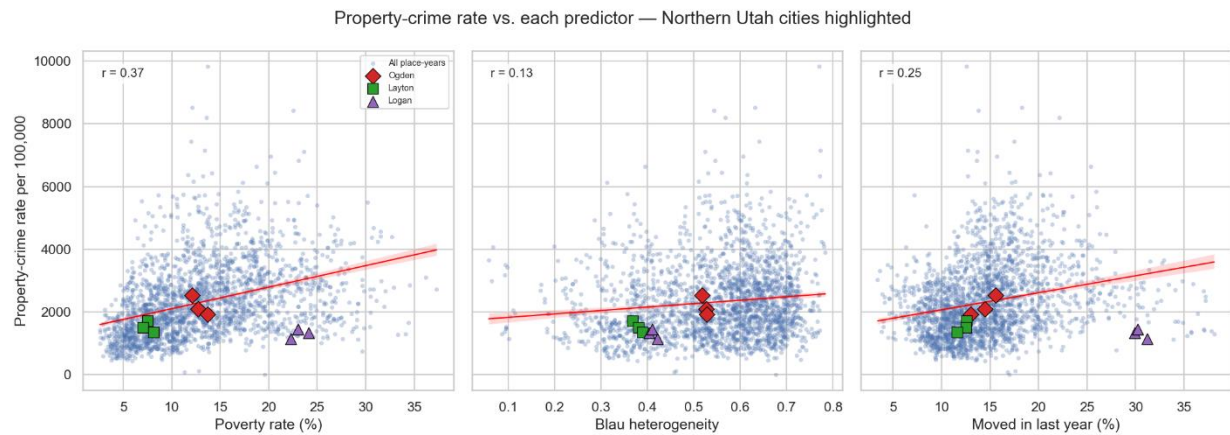


Figure 3. Bivariate scatter plots, with Northern Utah cities (Ogden, Layton, and Logan) highlighted.

Poverty rate shows the strongest linear association with the crime rate (Pearson $r = 0.37$). Residential instability ($r = 0.25$) and Blau heterogeneity ($r = 0.13$) show weaker but positive associations. No nonlinear patterns are visually apparent. The three Northern Utah focus cities are highlighted: Logan (triangles) sits far to the right on both the poverty and mobility panels, placing it among the highest-risk predictor values in the national sample. Ogden (diamonds) falls near the center of the cloud with its three years drifting right on poverty. Layton (squares) clusters in the low-poverty, low-mobility corner.

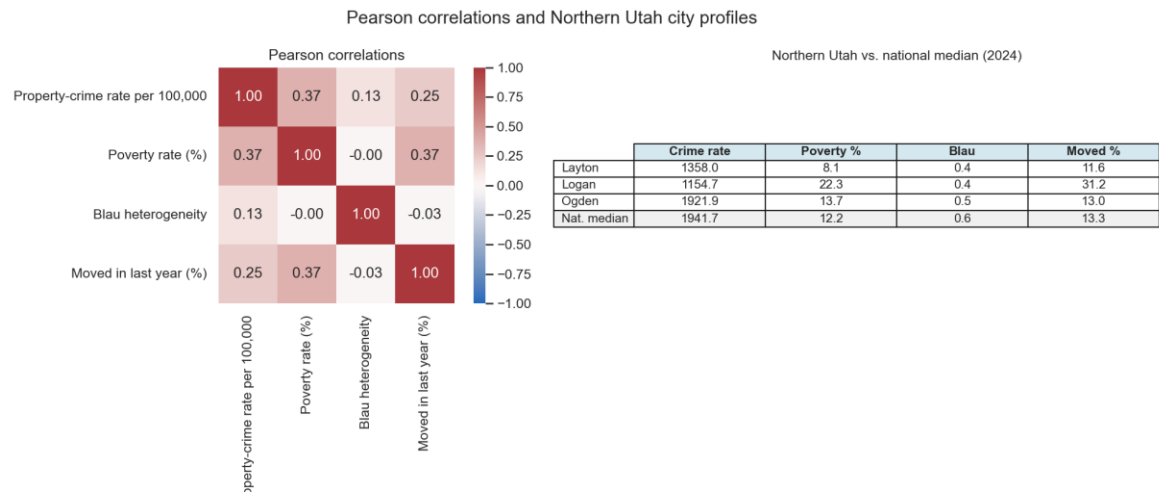


Figure 4. Pairwise Pearson correlation heatmap for the property crime rate and three predictors.

The heatmap confirms the predictors are only modestly related to one another, with the strongest pairing, poverty and mobility, still well short of any level that would raise concern. Stage 2 verifies this independence more formally with variance inflation factors.

All three predictors are statistically significant at $\alpha = 0.05$. Each additional percentage point of poverty is associated with approximately 60 more property crimes per 100,000 residents, holding the other predictors constant. A 0.1 increase in the Blau heterogeneity index (a large shift, since the index ranges from 0 to 1) corresponds to approximately 1,140 additional crimes per 100,000. Each additional percentage point of residents who moved in the past year is associated with approximately 28 more crimes per 100,000. The R^2 of 0.170 indicates that the three community conditions explain 17% of the variation in the property crime rate. Compared to a single-predictor baseline using foreign-born share ($R^2 = 0.021$), the theory model explains 8× more variance, confirming that the social-disorganization predictors substantially outperform the folk assumption. The remaining 83% of variation reflects other factors (policing, geography, economic structure) beyond the scope of this model.

The overall F-test is significant ($p < 0.001$) and all three coefficient p-values are below 0.05; H_0 is rejected at $\alpha = 0.05$. Poverty rate, ethnic heterogeneity, and residential instability statistically significantly affect the property crime rate.

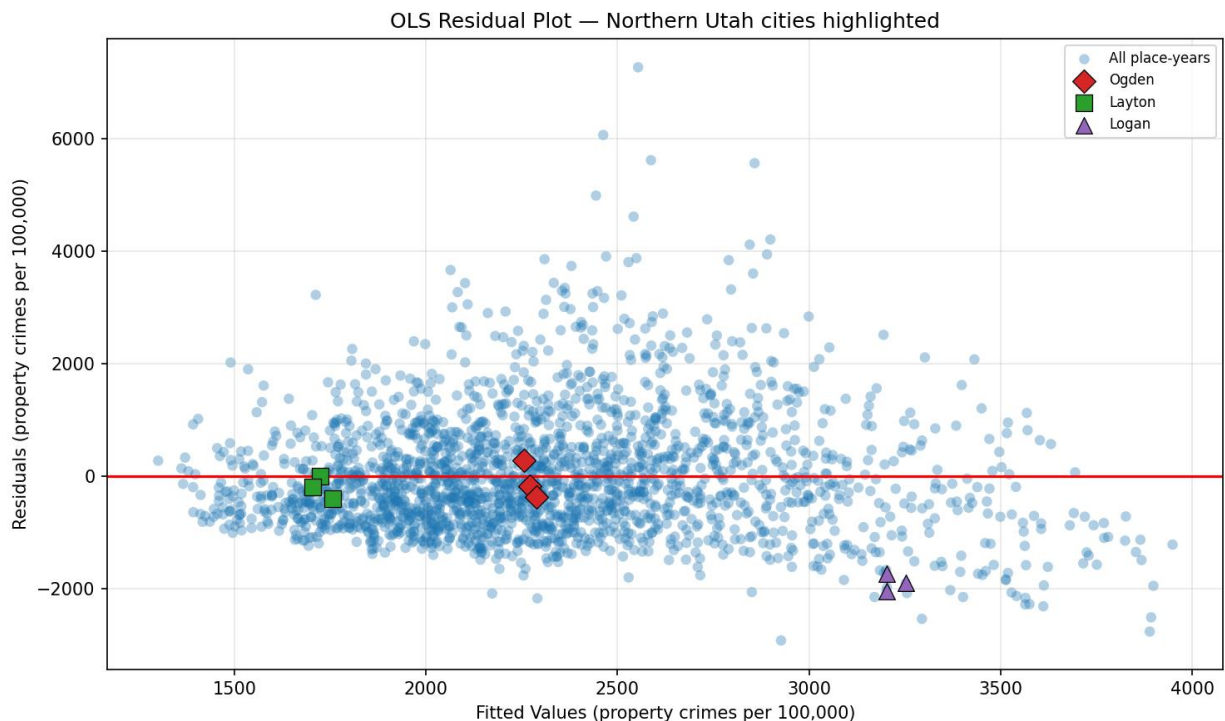


Figure 5. Residual vs. fitted plot, with Northern Utah cities highlighted.

The residual plot (Figure 5) shows a roughly horizontal band centered near zero, with increasing spread at higher fitted values. That widening fan is heteroscedasticity, which means the model's predictions tighten up and grow more reliable in the low-crime cities but scatter more widely, and are therefore less precise, for the high-crime cities. The three Northern Utah briefing cities are highlighted: Ogden (diamonds) sits near the center of the cloud with residuals close to zero, meaning the model predicts Ogden's crime rate reasonably well. Layton (squares) falls at the low end of fitted values with small residuals. Logan (triangles) sits at the high end of

fitted values; the model predicts a much higher crime rate for Logan's community conditions than what is actually recorded, producing large negative residuals (see Figure S1).

The Durbin–Watson statistic of 0.64 is well below the ideal value of 2, which signals autocorrelation: because the same 771 cities each appear in up to three consecutive years, a city's residual in one year tends to resemble its residual in the next rather than reading as a fresh, independent observation. Both patterns are noted as limitations but do not invalidate the significance test at $\alpha = 0.05$ for the purpose of rejecting H_0 .

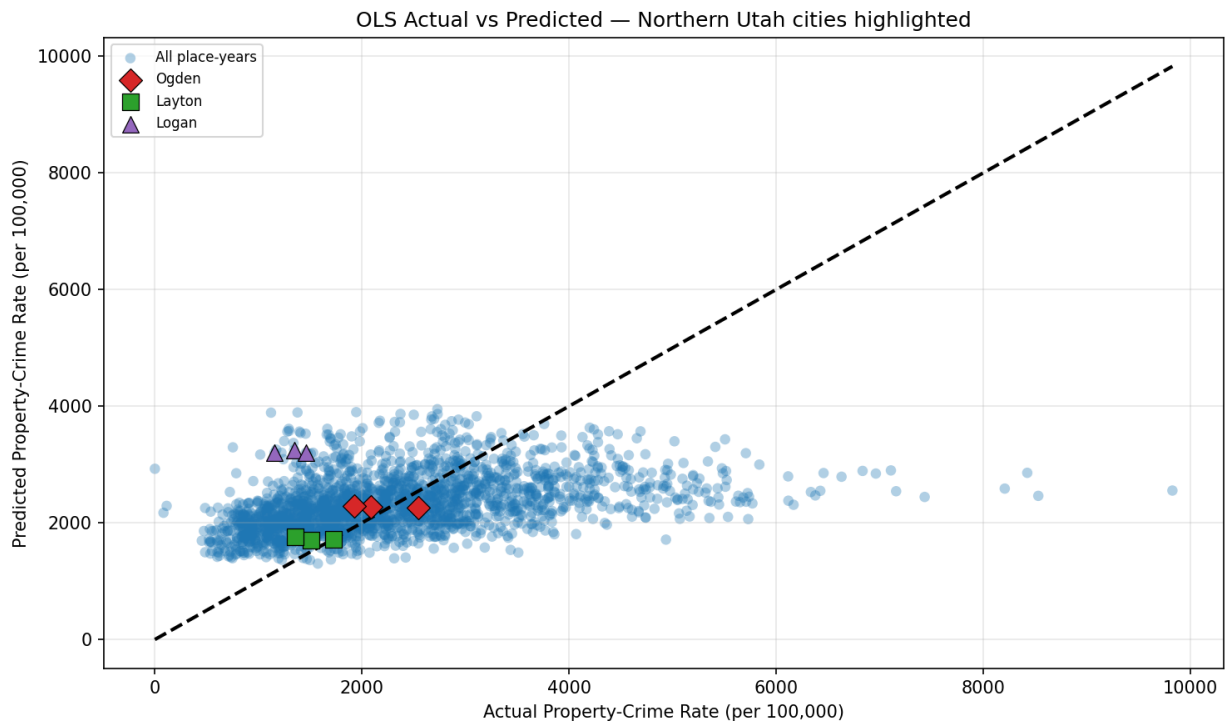


Figure S1 (supplementary). OLS actual vs. predicted property-crime rate, with Northern Utah cities highlighted.

All three VIFs are well below the flag threshold of 5, indicating that the predictors are essentially independent. The OLS coefficients are not destabilized by multicollinearity, and the individual p-values can be interpreted without concern that one predictor is masking another.

RidgeCV selected $\lambda = 26.37$, with training $R^2 = 0.185$ and standardized coefficients of 423.0 for poverty rate, 179.2 for Blau heterogeneity, and 110.9 for percent moved.

On the holdout set, Ridge achieved RMSE = 1,050 crimes per 100,000, compared to a mean-rate baseline RMSE of 1,021. The holdout RMSE indicates that Ridge predictions do not improve on simply predicting the training-set mean for the held-out divisions. This is not unexpected with only three predictors and a modest R^2 ; the value of Ridge here is coefficient stabilization for the LP, not standalone prediction accuracy.

After fitting, the analysis applied the Ridge model to every one of the 2,182 place-years to produce a predicted crime rate for each row, and confirmed that no prediction was negative, so none required clipping at zero. The predicted rates ranged from approximately 1,293 to 4,188

crimes per 100,000. These scores were then divided into fixed risk tiers, high (top 10%), mid (next 15%), and low (remaining 75%), which yielded 219 high-tier, 327 mid-tier, and 1,636 low-tier place-years. Those tiers carry the exposure into the linear program below.

Compared to an equal-share scenario (\$833,333), in which total residual loss is 9,215,673 expected-count units, the solved mix achieves residual loss of 9,207,526 expected-count units, an improvement of approximately 8,147 expected-count units. The solved mix concentrates spending where exposure is highest: surveillance and alarm in the high-risk tier, fortification in the mid tier. Authorization receives \$0 because the scored property crimes (burglary, larceny-theft, motor vehicle theft) are external-entry and lot problems, not credentialed-access problems.

Table 2. Sensitivity analysis: $\pm 25\%$ perturbation on each $\alpha(k)$.

Scenario	Residual	vs. Equal-Share	Mix Change?
Base	9,207,526	8,147	N/A
Fortification $\times 0.75$	9,211,693	6,064	No
Fortification $\times 1.25$	9,203,359	10,230	No
Surveillance $\times 0.75$	9,207,667	8,125	Yes, alarm up
Surveillance $\times 1.25$	9,207,359	8,195	No
Alarm $\times 0.75$	9,207,589	8,189	No
Alarm $\times 1.25$	9,207,443	8,125	Yes, alarm up
Authorization $\times 0.75$	9,207,526	8,199	No
Authorization $\times 1.25$	9,207,526	8,095	No

The solved mix remained stable across $\pm 25\%$ perturbations on each $\alpha(k)$ (Table 2). The mix shifted only in the surveillance–alarm boundary (from \$3.5M/\$1.5M to \$2.5M/\$2.5M) when surveillance effectiveness was reduced by 25% or alarm effectiveness was increased by 25%. Fortification's \$5M mid-tier allocation held across all scenarios. The residual-avoided figure ranged from 6,064 to 10,230 expected-count units.

Discussion

The null hypothesis stated that poverty rate, ethnic heterogeneity, and residential instability do not statistically significantly affect the property crime rate. The OLS regression rejected H_0 at $\alpha = 0.05$ ($F = 148.91$, $p < 0.001$). All three predictors were individually significant (all p -values < 0.001). The model explains 17% of the variation in property crime rates across 2,182 place-years, an 8 $\times$ improvement over a single-predictor baseline using foreign-born share ($R^2 = 0.021$), which confirms that the social-disorganization predictors substantially outperform the folk assumption.

In practical terms, each additional percentage point of poverty is associated with about 60 more property crimes per 100,000 residents. A one-unit increase in the Blau heterogeneity index corresponds to about 1,140 more crimes per 100,000, though because the index ranges from 0 to 1, real-world shifts are much smaller than one full unit. Each additional percentage point of residential mobility is associated with about 28 more crimes per 100,000. These findings

are consistent with social disorganization theory: communities with higher poverty, greater ethnic fragmentation, and more residential turnover exhibit higher property crime rates.

Ridge regression and the linear program translated these findings into an actionable security-budget recommendation, the optimal mix allocates 50% to fortification (physical barriers, fencing, lighting), 35% to surveillance (cameras and monitoring), 15% to alarm (intrusion detection), and 0% to authorization (access control), as shown in Figure 6. In practice, an owner applies the 50/35/15 percentages to a security budget sized from the fixed-asset investment the business already funds as a share of revenue. This solved mix avoided approximately 8,147 more expected-count units of residual loss than an equal-share baseline, concentrating resources where exposure is highest.

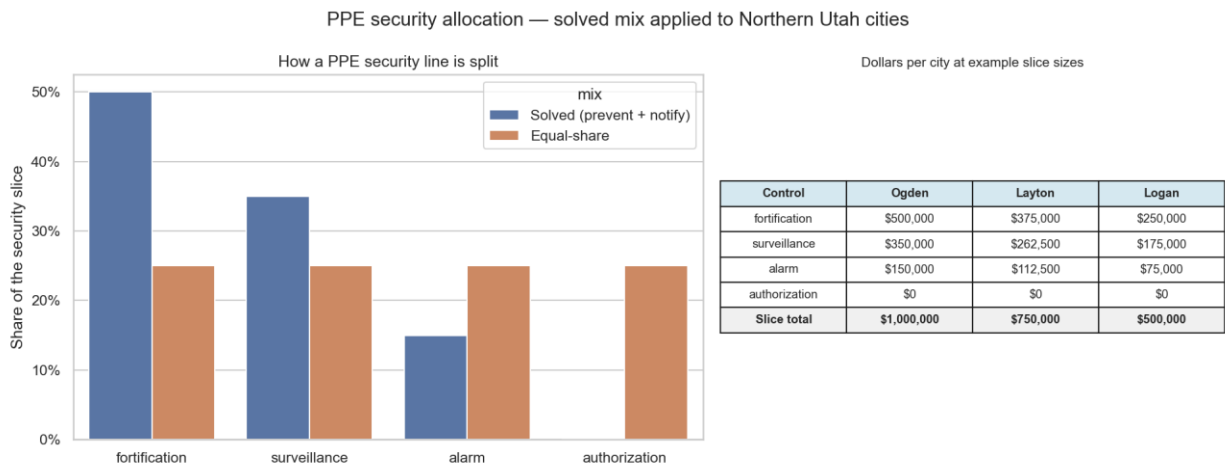


Figure 6. Recommended PPE security-slice allocation: solved mix vs. equal-share.

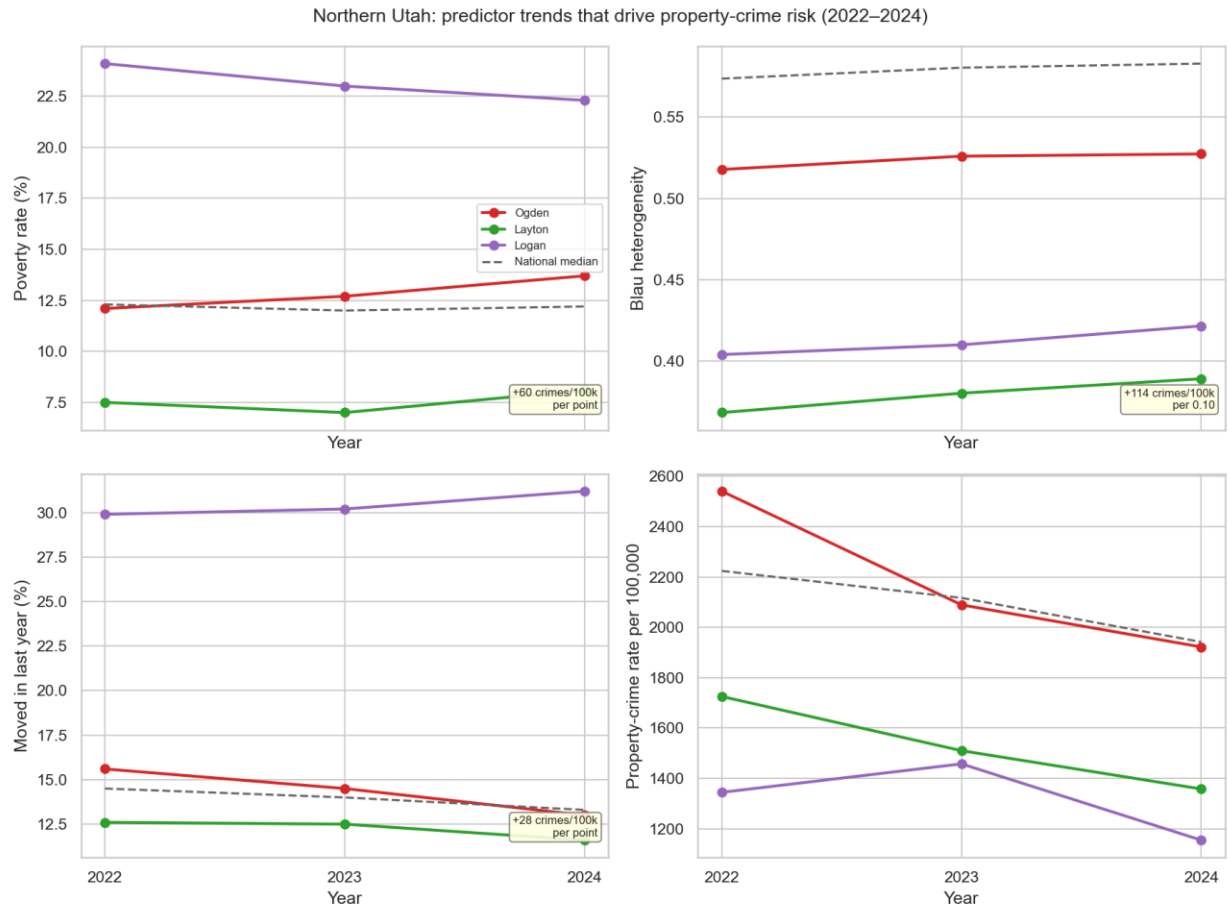


Figure 7. Northern Utah predictor trends (2022–2024), with coefficient impact annotations.

Ogden, Layton, and Logan are expanding cities: between 2022 and 2024, Layton grew 1.9% and Logan 3.1%, both well above the national median city growth of 0.4%. Expanding cities attract new plant, property, and equipment investment, which makes the property-crime question operationally relevant for owners evaluating sites. Table 3 shows both the raw offense count reported to the FBI and the rate per 100,000 residents; all three cities saw property crime fall between 2022 and 2024:

Table 3. Actual population and property crimes of Northern Utah focus cities across study years (2022–2024).

City	Pop. (2024)	2022	2023	2024	Change
Ogden	87,413	2,204 (2,541/100k)	1,817 (2,089/100k)	1,680 (1,922/100k)	–524 offenses (–25%)
Layton	83,286	1,410 (1,725/100k)	1,246 (1,510/100k)	1,131 (1,358/100k)	–279 offenses (–21%)
Logan	54,907	716	786	634	–82 offenses

		(1,345/100k)	(1,458/100k)	(1,155/100k)	(-11%)
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That is the good news; the question for a property owner is whether the decline will continue, and the answer depends on where the underlying risk drivers are heading.

Figure 7 showed the three predictors trending over the panel for each Northern Utah city, with the national median as a reference. The yellow annotations translate each predictor into its crime-rate impact using the OLS coefficients: each point of poverty adds about 60 crimes per 100,000; each point of mobility adds about 28; each 0.10 of the Blau index adds about 114.

In Ogden, reported property crimes dropped from 2,204 to 1,680 over three years, a drop of 524 burglaries, larcenies, and motor-vehicle thefts. The rate fell 25%, from 2,541 to 1,922 per 100,000, dropping below the national median by 2024. That is a meaningful improvement. But poverty climbed from 12.1% to 13.7%, crossing the national median. That 1.6-point increase, at the regression coefficient of ~60 crimes per 100,000 per point, translates to about 96 additional expected crimes per 100,000, or roughly 84 more offenses per year in a city of 87,000. Residential mobility fell (15.6% to 13.0%), offsetting about 73 expected crimes per 100,000 (~64 offenses). Net: the two forces nearly cancel. The recorded count is improving, but the strongest predictor (poverty) is moving the wrong way. Watch the poverty line.

In Logan, the city recorded 716 property crimes in 2022, 786 in 2023, and 634 in 2024, the smallest raw numbers of the three because Logan's population (~55,000) is about two-thirds of Ogden or Layton. The per-capita rate (1,155–1,458 per 100,000) is also the lowest in two of three years. On paper, Logan looks calm, but poverty sits above 22%, roughly 10 points above the national median, and about 30% of residents moved in the last year, more than double Layton's rate. Although we can understand this as a college town with a poverty rate and seasonal rotation of a percentage of its population being explained by students, the regression reads this as high risk: the poverty gap alone (Logan 22–24% vs. Layton 7–8%) implies ~900 more expected crimes per 100,000 than Layton, equivalent to about 500 additional offenses per year scaled to Logan's population. Ridge scores Logan high in all three years. The low recorded count may reflect local factors the model does not capture (e.g., campus policing), but the structural conditions that drive property crime nationally are elevated. An owner in Logan should not treat the low recorded count as a reason to underinvest in security.

In Layton, reported property crimes fell from 1,410 to 1,131, a 21% decline, 279 fewer offenses. The rate (1,358 per 100,000 in 2024) stayed consistently below the national median. Poverty stayed below 9%, mobility below 13%, Blau index around 0.38. All three predictors are below the national medians in every year. The regression confirms what the recorded count shows: Layton is the lowest-risk site. An owner here can maintain baseline security spending, but should re-check when new ACS estimates publish.

The R^2 of 0.170 means the three social-disorganization predictors account for about 17% of the variation in property-crime rates, which in turn means 83% of the variation comes from factors this model does not measure. A property owner should treat these results as one validated layer of a larger risk picture, not the whole picture.

The model addresses the foreign-born misconception with a quantified comparison, separates higher-risk from lower-risk cities on three measurable community conditions, and translates relative risk into a budget-allocation starting point.

The model does not address the remaining 83%, which reflects real forces that published research has identified:

- Policing intensity and strategy. Patrol density, hot-spot policing, and clearance rates directly affect offender calculus. A city with strong directed patrols may record less crime than its community conditions predict.²⁴
- Routine activities / opportunity. Property crime requires a motivated offender, a suitable target, and the absence of a capable guardian, all at the same time and place. Land use, business hours, foot traffic, and lighting shape opportunity independently of poverty or mobility.²⁵
- Collective efficacy. Neighborhoods where residents know and trust each other intervene informally against disorder. This may explain why Logan's recorded crime is lower than its profile predicts.²⁶
- Income inequality. The Gini coefficient within a city may matter more than the poverty rate alone. High inequality generates relative deprivation that motivates property offending even in cities with moderate average poverty.²⁷
- Built environment and land use. Street layout, mixed-use zoning, vacant lots, and transit access create or suppress crime opportunities at the micro level.²⁸
- Economic cycle. Unemployment and consumer-spending shifts affect both motivation to offend and the volume of targets.²⁹

Additionally, the Durbin–Watson statistic of 0.64 indicates positive autocorrelation from the panel structure (the same cities appear across three years), which means the OLS standard errors are likely too narrow. A more rigorous panel model (e.g., fixed-effects or random-effects) would be better accounted for within-city correlation.

For a Northern Utah property owner, this analysis answers a narrow but important question: do the social-disorganization conditions matter, and by how much? The answer is yes; they matter 8× more than the foreign-born shortcut, and each predictor has a quantified per-unit effect. But an owner making a site-level security decision should layer this analysis with:

- local police data from Ogden, Layton, and Logan PDs;

²⁴ (Weisburd et al., 2016)

²⁵ (Cohen & Felson, 1979)

²⁶ (Sampson et al., 1997)

²⁷ (Itskovich, 2024)

²⁸ (Lanfear, 2022)

²⁹ (Cantor & Land, 1985)

- a site-specific CPTED audit to determine where on the property to place controls;
- insurance loss history, which captures unreported losses FBI data miss; and
- updated ACS releases, which refresh the predictors annually.

None of these layers invalidates the regression; they complement it. The 17% this model explains is the community-condition layer. The other layers fill different parts of the remaining 83%.

Because H_0 was rejected, the three community conditions provide a defensible basis for security-budget allocation. PPE owners in cities with higher poverty, higher ethnic heterogeneity, and higher residential instability should allocate a greater share of their security budget to preventive controls. Specifically, the linear program recommends a security-slice split of 50% fortification (delay), 35% surveillance (watch), and 15% alarm (notify), with zero allocation to authorization (access control). For a concrete example: an owner with a \$1 million security slice in Ogden, Utah, would allocate \$500,000 to fencing, lighting, and physical hardening; \$350,000 to camera systems and monitoring; and \$150,000 to intrusion-detection alarms. Sites in high-predictor cities like Logan need a larger absolute security slice than low-predictor cities like Layton, even though the percentage mix stays the same.

An owner should re-assess when updated ACS estimates publish. The single most important trend is the poverty rate; it is the strongest predictor ($r = 0.37$, $\beta \approx 60$ crimes per 100,000 per point) and the one rising in Ogden. If Ogden's poverty rate crosses 15%, the expected crime-rate increase (about 78 additional crimes per 100,000 from the 2024 baseline) would warrant increasing the absolute dollar amount of the security slice even if the percentage mix stays at 50/35/15. Residential mobility is the second lever: each percentage-point decline offsets about 28 expected crimes. As long as mobility is falling, it partially buffers a rising poverty rate, but the buffer is smaller per unit than the poverty effect.

- Routine-activities layer. Adding land-use mix (e.g., commercial-to-residential ratio from Census NAICS data) and a guardianship proxy (e.g., law-enforcement employees per capita from FBI LEOKA tables) would test whether opportunity-theory variables explain incremental variance beyond social disorganization.
- Collective-efficacy survey. For the three Northern Utah cities specifically, a short resident survey measuring mutual trust and willingness to intervene would capture the informal-control mechanism that ACS demographics cannot proxy, and help explain why Logan's recorded crime is lower than its predictor profile predicts.
- Income-inequality (Gini) predictor. The ACS publishes a Gini coefficient at the place level (table B19083). Adding it as a fourth predictor would test whether inequality explains variance beyond the poverty rate alone.
- NIBRS incident-level disaggregation. Using National Incident-Based Reporting System microdata, which include offense type, location type, and time of day, would allow the LP to differentiate between after-hours warehouse burglaries (fortification + alarm) and daytime retail larceny (surveillance), sharpening the security-mix recommendation.
- Spatial lag / spatial error models. Property crime rates in neighboring cities are not independent. A spatial regression would account for cross-boundary spillover, which is

relevant in Northern Utah, where Ogden, Layton, and Logan sit along the same I-15 corridor.³⁰

Each of these would add a piece to the 83% of variance this model leaves on the table. The current study's contribution is to establish the community-condition baseline and to retire the foreign-born shortcut with a quantified comparison.

Declarations

Data availability. All data are publicly available from the FBI Crime Data Explorer, the U.S. Census Bureau American Community Survey and Gazetteer files, and the Bureau of Justice Statistics National Crime Victimization Survey.

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³⁰ (Anselin, 2003)

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